1. Explain Shor's and Grover's contributions to quantum computing. How do they demonstrate the power of quantum algorithms?

- **Shor's Algorithm (1994):**
  - Provides a polynomial-time solution to the integer factorization problem.
  - This was significant because classical algorithms for factoring are super-polynomial in time.
  - The algorithm threatens the security of RSA cryptosystem, which relies on hardness of factoring.
- **Grover's Algorithm (1996):**
  - Provides a quadratic speedup for unstructured search.
  - While a classical search takes O(N) steps, Grover's algorithm finds the solution in $O(\sqrt{N})$ steps.
- **Significance:**
  - Shor's algorithm shows exponential speedup for certain problems.
  - Grover's algorithm shows general-purpose quadratic improvement.
  - Together, they demonstrated that quantum computers can outperform classical ones for specific problems.

2. Describe the photon polarization experiment with polaroids. How does quantum mechanics explain the surprising result?

- **Experiment Steps:**
  1. Shine unpolarized light through polaroid A → only 50% passes (horizontal polarization).
  2. Place polaroid C orthogonal (vertical) to A → no light passes.
  3. Insert polaroid B at 45° between A and C → surprisingly, some light passes through C.
- **Quantum Explanation:**
  - Photons are modeled as polarization states (unit vectors).
  - After polaroid A, photons are in $|\rightarrow\rangle$ (horizontal).
  - When polaroid B ($|\nearrow\rangle$ axis) is introduced, photons have a 50% chance to pass and collapse into $|\nearrow\rangle$.
  - In this state, they have amplitude in the vertical direction, so some pass through C.
- **Conclusion:**
  - Quantum mechanics explains the non-intuitive phenomenon using **superposition and probabilistic measurement**.

3. Discuss the differences between classical bits and quantum bits (qubits) with examples.

- **Classical Bit:**
  - Can take values 0 or 1.
  - Represented by physical systems with two distinct states (e.g., high/low voltage).
- **Quantum Bit (Qubit):**
  - Represented as $|\psi\rangle = a|0\rangle + b|1\rangle$, where $|a|^2+|b|^2=1$.
  - Can exist in **superpositions** of 0 and 1.

    - Measurement collapses it to either $|0\rangle$ or $|1\rangle$ probabilistically.
- **Examples:**
    - Photon polarization ($|\uparrow\rangle$, $|\rightarrow\rangle$).
    - Electron spin (spin-up, spin-down).
- **Key Difference:**
    - Bits are discrete and deterministic.
    - Qubits have infinite possible states and exhibit quantum phenomena like superposition and entanglement.

4. Explain the concept of qubit measurement. Why does measurement limit the information we can extract from a qubit?

- **Measurement Process:**
    - A qubit $|\psi\rangle = a|0\rangle + b|1\rangle$ is measured with respect to a basis (usually $\{|0\rangle, |1\rangle\}$).
    - Result:
        - Collapses to $|0\rangle$ with probability $|a|^2$.
        - Collapses to $|1\rangle$ with probability $|b|^2$.
- **Effects:**
    - Measurement **changes the state** irreversibly (collapse).
    - Subsequent measurements in the same basis yield the same outcome deterministically.
- **Limitation on Information:**
    - Though qubits have infinitely many possible superpositions, measurement yields only **one classical bit** of information.
    - Cannot measure in multiple bases simultaneously.
    - **No-cloning theorem** prevents making copies for different measurements.
- **Conclusion:**
    - Measurement defines the boundary between quantum potential and classical reality, limiting extracted information.

5. Discuss the importance of the no-cloning principle and its implications in quantum information processing.

- **No-Cloning Theorem:**
    - States that it is impossible to perfectly copy an unknown quantum state.
    - Unlike classical data, quantum states cannot be duplicated without disturbing them.
- **Reason:**
    - Quantum states exist in superpositions.
    - Copying would require duplicating amplitudes for all possible states, which is mathematically impossible under unitary evolution.
- **Implications:**
    - Ensures security in quantum cryptography (e.g., eavesdropper cannot copy qubits undetected).
    - Limits error correction—quantum error correction relies on entanglement and redundancy, not duplication.
    - Shapes the design of quantum communication protocols.
- **Significance:**

- No-cloning is a fundamental property of quantum mechanics that underlies the **security and uniqueness** of quantum information.

6. Explain the concept of a qubit with an example.

A **qubit** is the fundamental unit of quantum information. Unlike a classical bit that takes only values 0 or 1, a qubit can exist in a **superposition**:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$.

For example, in photon polarization, |0⟩ can represent vertical polarization (|↑⟩) and |1⟩ horizontal polarization (|→⟩). A photon polarized at 45° corresponds to the state

$$|\nearrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

7. Describe the role of measurement in a single-qubit system.

Measurement is a **fundamental postulate of quantum mechanics**.

- A measuring device has two preferred states forming an **orthonormal basis** (e.g., {|0⟩, |1⟩}).
- When a qubit in state $|\psi\rangle = a|0\rangle + b|1\rangle$ is measured:
  - It collapses to |0⟩ with probability $|a|^2$.
  - It collapses to |1⟩ with probability $|b|^2$.
- Measurement is **probabilistic** and **changes the state** irreversibly.
- Only **one classical bit of information** can be extracted from a qubit, regardless of its superposition.

8. Write short notes on Dirac's bra–ket notation.

- Introduced by Paul Dirac, **bra–ket notation** is the standard formalism in quantum mechanics.
- A **ket** |v⟩ represents a state vector (column vector).
- A **bra** ⟨v| is the conjugate transpose of the ket (row vector).
- The **inner product** ⟨u|v⟩ gives a scalar (complex number).
- Two states are **orthogonal** if ⟨u|v⟩ = 0.
- Orthonormal bases satisfy ⟨βi|βj⟩ = δij.

  Example:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

This notation simplifies representation of quantum states, measurements, and transformations.

9. Explain the BB84 quantum key distribution protocol.

The **BB84 protocol**, proposed by Bennett and Brassard in 1984, is the first quantum key distribution (QKD) protocol.

- **Setup:** Alice and Bob communicate via a classical channel and a quantum channel. Alice sends photons encoded as qubits.
- **Encoding:** Each bit is randomly encoded using either:
  - Standard basis: $0 \to |\uparrow\rangle$, $1 \to |\rightarrow\rangle$
  - Hadamard basis: $0 \to |\nearrow\rangle$, $1 \to |\nwarrow\rangle$
- **Measurement:** Bob randomly chooses a basis to measure. If his basis matches Alice's, he gets the correct bit.
- **Key Selection:** They discard mismatched basis results. The remaining bits form a shared key.
- **Eavesdropping Detection:** If Eve intercepts and measures qubits, she introduces ~25% errors, which Alice and Bob can detect.
  Thus, BB84 provides **secure key sharing** using quantum principles.

10. Distinguish between relative phase and global phase in qubits.

- **Global Phase:** Multiplying a quantum state by a complex number of modulus one (e.g., $e^{i\varphi}$) does not change the physical state.
  Example: $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ represent the same state.
- **Relative Phase:** The phase difference between components of a superposition is **physically significant**.
  Example:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{vs.} \quad \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

  These are different states with different measurement outcomes.
  Hence, **global phase is irrelevant**, while **relative phase determines interference and measurement probabilities**.

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11. Explain why only one bit of classical information can be extracted from a qubit.

- A qubit can exist in infinitely many states (superpositions).
- However, **measurement collapses** the qubit into one of two basis states.
- This collapse yields only **one outcome** ($|0\rangle$ or $|1\rangle$).
- The probabilistic nature of measurement prevents extracting more information.
- Moreover, due to the **no-cloning principle**, it is impossible to make copies of a qubit to measure in different bases.
  Thus, despite its infinite possibilities, a single qubit provides only **one classical bit of information** upon measurement.

12. Explain the difference between direct sums and tensor products of vector spaces with examples.

- **Direct Sum ($V \oplus W$):**
  - Represents systems where the state is either in V or in W.
  - Dimension adds: $\dim(V \oplus W) = \dim(V) + \dim(W)$.
  - Example: A coin (2 states) or a die (6 states) → $2 + 6 = 8$.
- **Tensor Product ($V \otimes W$):**
  - Represents quantum systems where subsystems combine.
  - Dimension multiplies: $\dim(V \otimes W) = \dim(V) \times \dim(W)$.
  - Example: Two qubits ($2 \times 2 = 4$ states).

13. Write and explain the general form of an n-qubit state

- An n-qubit system has dimension 2^n.
- The general state is:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle$$

  where $\alpha_i \in \mathbb{C}$, and $\sum |\alpha_i|^2 = 1$.
- Each basis state $|i\rangle$ corresponds to an n-bit binary string.
- Example (3 qubits):

$$|\psi\rangle = a|000\rangle + b|001\rangle + \cdots + h|111\rangle$$

- **Key Point:** Some states can be written as product states; others are **entangled**.

14. Define entangled states. Give examples of Bell, GHZ, and W states.

- **Entangled state:** A state that cannot be expressed as a product of individual qubit states.
- **Examples:**
  - **Bell states (2-qubit):**
    $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
    $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$
  - **GHZ state (3-qubit):**
    $|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$
  - **W state (3-qubit):**
    $|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$
- **Importance:** Entanglement enables quantum teleportation, superdense coding, and secure communication.

15. Explain the process and consequences of measuring part of a multi-qubit system.

- Measurement projects the state into a subspace, collapsing probabilities.
- Example:
  For 2-qubit state $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$:
  - Measuring first qubit:
    - Outcome 0: probability $|a|^2 + |b|^2$, state becomes $\frac{a|00\rangle + b|01\rangle}{\sqrt{|a|^2+|b|^2}}$.
    - Outcome 1: probability $|c|^2 + |d|^2$, state becomes $\frac{c|10\rangle + d|11\rangle}{\sqrt{|c|^2+|d|^2}}$.
- **Key features:**
  - Probabilistic outcomes.
  - Collapse of state.
  - Remaining qubits are left in a **conditional state**.

16. Describe Ekert's entanglement-based QKD protocol (E91).

- Proposed by **Artur Ekert (1991)**.
- Uses **entangled qubit pairs (EPR pairs)** shared between Alice and Bob.
- Steps:
  1. Entangled qubits distributed to Alice and Bob.
  2. Each randomly chooses measurement bases.
  3. Outcomes where bases match → raw key.
  4. Check correlations using **Bell's inequality** to detect eavesdropping.
  5. Apply error correction + privacy amplification → final key.
- **Security:**
  - Eve's interference breaks entanglement and introduces errors.
  - Violations of Bell's inequality guarantee secure correlations.

17. Explain the development of quantum computing from its origins to Shor's and Grover's algorithms. Why was this shift from classical to quantum computation significant?

Answer:

- **Origins:**
  - Quantum computing arose in the 1980s when scientists like Feynman and Manin realized that certain quantum phenomena could not be efficiently simulated by a classical Turing machine.
  - Charles Bennett and Gilles Brassard introduced **quantum cryptography** (BB84 protocol) exploiting quantum measurement.
- **Models of Computation:**
  - David Deutsch formalized the **quantum Turing machine**.
  - The **quantum circuit model** with **quantum gates** became the standard.
- **Key Algorithms:**
  - **Shor's Algorithm (1994):** Provided a polynomial-time solution to integer factorization, threatening RSA security.
  - **Grover's Algorithm (1996):** Provided quadratic speedup for unstructured search, reducing time from O(N) to O(√N).
- **Significance:**
  - These algorithms proved quantum computers can outperform classical ones for specific problems.
  - Shifted computing from a purely classical mechanical model to one grounded in **quantum mechanics**, with applications in cryptography, communication, and algorithms.
- **Conclusion:**
  - The shift was significant because it redefined the **concept of computation**, introducing superposition, entanglement, and measurement as computational resources.

18. Discuss in detail the photon polarization experiment with polaroids. How does quantum mechanics, through superposition and measurement, explain the results that seem impossible in classical physics?

- **Experiment Setup:**
  - Shine unpolarized light on a screen.
  - Place polaroid A (horizontal) → 50% light passes.
  - Add polaroid C (vertical, orthogonal to A) → no light passes.
  - Insert polaroid B (45° between A and C) → some light surprisingly passes.
- **Classical Expectation:**
  - Adding more filters should only reduce light further, never increase it.
- **Quantum Explanation:**
  - Polarization states are represented by unit vectors ($|\uparrow\rangle$, $|\rightarrow\rangle$).
  - After polaroid A, photons are in $|\rightarrow\rangle$.
  - With polaroid B at 45°, $|\rightarrow\rangle$ can be written as $(1/\sqrt{2})(|\nearrow\rangle - |\nwarrow\rangle)$.
  - This gives photons a non-zero chance of aligning with C's axis (vertical), so some pass through.

- **Key Concepts:**
  - **Superposition:** $|\rightarrow\rangle$ expressed as a combination of $|\nearrow\rangle$ and $|\nwarrow\rangle$.
  - **Measurement:** Probabilistic collapse into one basis direction, changing photon polarization.
- **Conclusion:**
  - The experiment highlights that quantum systems don't behave like classical sieves. Instead, superposition and measurement explain outcomes that seem impossible in classical physics.

19. Compare and contrast classical bits and qubits. Explain why qubits are more powerful, and also describe the limitations of extracting information from them.

Answer:

- **Classical Bits:**
  - Take discrete values 0 or 1.
  - Physical realizations include voltages (high/low), switches, etc.
- **Quantum Bits (Qubits):**
  - Represented as $|\psi\rangle = a|0\rangle + b|1\rangle$, with $|a|^2+|b|^2=1$.
  - Can exist in superpositions, allowing parallelism.
  - Examples: photon polarization, electron spin.
- **Power of Qubits:**
  - Superposition allows representation of multiple states simultaneously.
  - Entanglement enables correlations with no classical counterpart.
  - Foundation for speedups in algorithms like Shor's and Grover's.
- **Limitations:**
  - Measurement collapses qubits to basis states, yielding only **one bit of classical information**.
  - **No-cloning theorem** prevents copying for multiple measurements.
  - Quantum states are fragile and prone to **decoherence**.
- **Conclusion:**
  - Qubits provide vast computational potential, but measurement restrictions and fragility limit the extraction of information and practical use.

20. Explain the process of single-qubit measurement in detail. Why is measurement considered the bridge between quantum and classical information?

- **Measurement Postulate:**
  - Any measuring device for a qubit has a basis ($|u\rangle$, $|u\perp\rangle$).
  - A qubit $|\psi\rangle = a|u\rangle + b|u\perp\rangle$ collapses into $|u\rangle$ with probability $|a|^2$ or into $|u\perp\rangle$ with probability $|b|^2$.
- **Consequences:**
  - Measurement **changes the state** irreversibly.
  - A second measurement in the same basis always returns the same result.
  - Different bases give different probabilistic outcomes.
- **Superposition and Measurement:**
  - A state may appear deterministic in one basis but probabilistic in another.

    - Example: $|\nearrow\rangle$ gives a certain result in the Hadamard basis but random results in the standard basis.
- **Bridge to Classical:**
    - Qubits exist in infinitely many states.
    - Measurement reduces this quantum richness into a discrete classical outcome (0 or 1).
- **Conclusion:**
    - Measurement connects the quantum world of probabilities and superpositions to the classical world of definite outcomes, enabling quantum results to be observed.

21. Compare and contrast classical bits and quantum bits (qubits) with suitable examples.

- **Classical Bits:**
    - A bit can only take two values: 0 or 1.
    - Example: Two voltage levels in circuits, ON/OFF in switches.
    - Information is stored in a binary deterministic form.
- **Quantum Bits (Qubits):**
    - A qubit can be in states $|0\rangle$, $|1\rangle$, or any superposition:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1$$

    - Infinite possible states due to superposition.
    - Example: Photon polarization — vertical ($|\uparrow\rangle = |0\rangle$), horizontal ($|\rightarrow\rangle = |1\rangle$), or diagonal ($|\nearrow\rangle$).
    - Measurement collapses qubit probabilistically.
- **Key Differences:**
    1. Bits are deterministic, qubits probabilistic.
    2. Bits cannot exist in superposition; qubits can.
    3. Bits give full information; qubits yield at most 1 classical bit on measurement.
    4. Qubits support entanglement, a non-classical property.

Thus, qubits provide richer information representation, enabling phenomena like quantum parallelism, unlike classical bits.

22. Describe the BB84 protocol in detail and explain how eavesdropping is detected.

- **Purpose:** BB84 (Bennett–Brassard, 1984) is a **quantum key distribution protocol** to securely share a random key.
- **Steps:**
    1. **Preparation:** Alice generates random bits and encodes them into photon polarizations using two bases:
        - Standard basis: $0 \rightarrow |\uparrow\rangle$, $1 \rightarrow |\rightarrow\rangle$
        - Hadamard basis: $0 \rightarrow |\nearrow\rangle$, $1 \rightarrow |\nwarrow\rangle$
    2. **Transmission:** Alice sends photons (qubits) over the quantum channel to Bob.
    3. **Measurement:** Bob randomly chooses a basis for measurement.
    4. **Basis Comparison:** Over the classical channel, Alice and Bob compare bases (not values). Mismatched bases are discarded.

5. **Key Extraction:** The remaining bits form the **raw key**.
6. **Error Checking:** They publicly compare a subset of bits. If the error rate is high, eavesdropping is detected.

- **Eavesdropping (Eve's Attack):**
  - Eve must measure before resending to Bob.
  - If she chooses the wrong basis (probability 50%), she alters the photon state.
  - This introduces about **25% error rate** in Bob's results.
  - Alice and Bob detect this error during verification, ensuring security.

23. Explain with examples the concept of superposition and its role in quantum measurement.

- **Definition:**

  A superposition is a linear combination of basis states. For a qubit:

  $$|\psi\rangle = a|0\rangle + b|1\rangle$$

  where $|a|^2 + |b|^2 = 1$.
- **Example:**
  - $|0\rangle$ = vertical polarization $|\uparrow\rangle$, $|1\rangle$ = horizontal polarization $|\rightarrow\rangle$.
  - A photon at 45° is in state:

  $$|\nearrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$
- **Measurement in Standard Basis:**
  - Probability of measuring $|0\rangle$ = $|a|^2$.
  - Probability of measuring $|1\rangle$ = $|b|^2$.
  - Example: $|\nearrow\rangle$ gives 50% chance of $|0\rangle$ and 50% of $|1\rangle$.
- **Significance:**
  - Superposition allows quantum parallelism (simultaneous representation of multiple states).
  - But measurement collapses it to a single outcome.
  - Distinguishes quantum systems from classical randomness.

24. Discuss the importance of relative phase versus global phase in quantum states.

- **Global Phase:**
  - A state multiplied by $e^{i\varphi}$ (φ real) represents the same physical state.
  - Example: |ψ⟩ and $e^{i\pi}|\psi\rangle = -|\psi\rangle$ are indistinguishable.
  - Has no observable effect.
- **Relative Phase:**
  - Phase difference between components of superposition is significant.
  - Example:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{vs.} \quad \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

  - Though similar in magnitude, they yield **different interference outcomes** in experiments.
- **Role in Measurement:**
  - Relative phase determines probabilities in certain measurement bases.
  - Example: In Hadamard basis, |0⟩ + |1⟩ collapses deterministically to |↗⟩, while |0⟩ − |1⟩ collapses to |↖⟩.
- **Conclusion:**
  - **Global phase**: irrelevant, unobservable.
  - **Relative phase**: essential, governs interference and outcomes.

25. Explain why measurement limits the amount of information that can be extracted from a qubit.

- A qubit can exist in infinitely many states (superpositions).
- However, measurement in a chosen basis collapses the state to **one of two eigenstates**.
- Thus, each measurement yields **only 1 classical bit of information**.
- Example: A state |ψ⟩ = (1/√2)(|0⟩ + |1⟩) gives a random result of 0 or 1 on measurement.
- **No-Cloning Principle:** Prevents making copies of the qubit to measure in multiple bases.
- Therefore, it is **impossible to fully determine an unknown qubit state** from measurement.
- This limitation is exploited in quantum cryptography (e.g., BB84) for security.

26. Explain the structure of the state space of an nnn-qubit system and discuss the significance of entanglement.

- The state space of an n-qubit system is the tensor product of n single-qubit spaces:

$$(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$$

- Basis = all n-bit binary strings.
- General state:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle, \quad \sum |\alpha_i|^2 = 1$$

- For 2 qubits: $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$.
- **Entanglement:**
  - Most states cannot be expressed as a product of single-qubit states.
  - Example: $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.
  - Significance: Entanglement enables non-classical correlations, quantum teleportation, superdense coding, QKD, and speedups in quantum algorithms.

27. Discuss the Bell states in detail and explain their importance in quantum communication.

- Bell states are maximally entangled 2-qubit states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\backslash\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\backslash\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\backslash\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

- Properties:
  - Maximally correlated measurement outcomes.
  - Form an **orthonormal basis** for 2-qubit systems.
- Importance in communication:
  - Used in **quantum teleportation** (transfer of unknown states).
  - Used in **superdense coding** (send 2 classical bits with 1 qubit).
  - Basis for **QKD** security in Ekert's protocol.
- Bell states demonstrate the non-classical feature of **entanglement**, essential for quantum information.

28. Explain with an example how measuring part of an entangled system produces correlations between outcomes.

- Consider Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- If Alice measures her qubit:
  - If she gets $|0\rangle$, Bob's qubit collapses to $|0\rangle$.
  - If she gets $|1\rangle$, Bob's qubit collapses to $|1\rangle$.
- Probabilities:
  - Each outcome occurs with probability $1/2$.
- Key Feature:
  - Outcomes are **perfectly correlated**, even at a distance.
  - No classical explanation — shows **non-local correlations**.
- Significance:
  - Basis of secure communication (QKD).
  - Demonstrates difference between quantum and classical information.

29. Compare BB84 and Ekert's E91 protocol for quantum key distribution.

Feature	BB84 (1984)	Ekert E91 (1991)
Resource used	Single qubits	Entangled qubit pairs
Security principle	No-cloning theorem + random bases	Violation of Bell's inequality
Implementation	Simpler, widely deployed	More complex, requires entanglement
Detection of Eve	Error rates in key	Breaking of entanglement correlations
Advantage	Practical for near-term systems	Stronger theoretical guarantees

BB84: Practical and widely implemented.
E91: Provides security from fundamental quantum correlations, even against powerful adversaries.

30. Describe in detail the differences between classical computation and quantum computation. How do qubits, superposition, and entanglement redefine the model of computation?

- **Classical Computation:**
  - Based on **bits** taking values 0 or 1.
  - Follows deterministic rules; Turing machine is the standard model.
  - Parallelism requires multiple processors.
- **Quantum Computation:**
  - Based on **qubits**, which can exist in states $|0\rangle$, $|1\rangle$, or a superposition $a|0\rangle+b|1\rangle$.
  - **Superposition:** Allows qubits to represent multiple values simultaneously.
  - **Entanglement:** Links qubits in a way that measurements on one affect the state of another, no matter the distance.
  - **Quantum gates:** Unitary transformations that evolve qubit states.
- **Implications:**
  - Algorithms can explore multiple possibilities at once.
  - Quantum algorithms (Shor's, Grover's) outperform classical counterparts.
  - Redefines computation from deterministic step-by-step to **probabilistic, parallel, and reversible transformations**.
- **Conclusion:**
  - Quantum computation represents a paradigm shift, offering new power but also new challenges such as decoherence and error correction.

31. Explain the photon polarization experiment with polaroids as an illustration of quantum mechanics. Discuss its importance in defining qubits and the principles of measurement.

- **Setup:**
  - Shine light on a screen.
  - Insert polaroid A (horizontal) → half intensity passes.
  - Insert polaroid C (vertical, orthogonal to A) → no light passes.

    - Insert polaroid B (at 45° between A and C) → surprisingly, some light passes.
- **Classical Expectation:**
    - Additional filters should further block light.
- **Quantum Explanation:**
    - Photon polarization is represented by states $|\uparrow\rangle$ and $|\rightarrow\rangle$.
    - After A: photons are in $|\rightarrow\rangle$.
    - Express $|\rightarrow\rangle$ in terms of $|\nearrow\rangle$ and $|\nwarrow\rangle$ (superposition).
    - With B inserted, photons collapse into $|\nearrow\rangle$ with probability ½, allowing some to pass C.
- **Importance:**
    - Demonstrates **superposition** and **basis-dependent measurement**.
    - Shows how probabilities arise from amplitudes.
    - Provides a physical basis for defining **qubits** using polarization states.
- **Conclusion:**
    - This experiment is a simple but powerful demonstration of how quantum mechanics departs from classical intuition and motivates quantum information concepts.

32. Discuss Shor's algorithm and its implications for cryptography. Why did it mark a turning point in quantum computing research?

- **Problem Context:**
    - Many cryptographic systems (RSA) rely on the hardness of factoring large integers.
    - Classical algorithms for factoring require super-polynomial time.
- **Shor's Contribution (1994):**
    - Developed a polynomial-time quantum algorithm for integer factorization.
    - Uses **Quantum Fourier Transform** to extract periodicity in modular arithmetic functions.
- **Implications:**
    - Threatens RSA, which secures much of today's digital communication.
    - Showed that quantum computers could solve practically important problems, not just theoretical ones.
- **Turning Point:**
    - Sparked intense research in quantum computing.
    - Motivated efforts in quantum hardware, quantum error correction, and new algorithms.
    - Demonstrated clear separation of computational power between classical and quantum models.
- **Conclusion:**
    - Shor's algorithm remains one of the strongest justifications for quantum computing research, linking theory with real-world impact.

33. Explain the process of single-qubit measurement. Discuss its probabilistic nature, the collapse of the wave function, and why it limits extraction of information from qubits.

- **Measurement Rule:**
    - A qubit $|\psi\rangle = a|0\rangle + b|1\rangle$ is measured in the standard basis $\{|0\rangle, |1\rangle\}$.

    - Probability($|0\rangle$) = $|a|^2$; Probability($|1\rangle$) = $|b|^2$.
- **Collapse of the Wavefunction:**
    - Measurement changes the qubit to the observed basis state.
    - Repeated measurement gives the same result deterministically.
- **Probabilistic Nature:**
    - Unlike classical systems, outcomes cannot be predicted deterministically, only probabilistically.
    - Example: $|\nearrow\rangle = (1/\sqrt{2})(|0\rangle+|1\rangle)$ yields 50% $|0\rangle$ and 50% $|1\rangle$.
- **Limitations:**
    - Infinite possible qubit states collapse into only two classical outcomes.
    - Measurement yields only **one bit of classical information**.
    - No-cloning theorem prevents multiple independent measurements.
- **Conclusion:**
    - Measurement bridges the quantum and classical worlds, but fundamentally restricts how much information can be extracted from quantum states.

34. Explain in detail the process of quantum measurement for a single qubit. Why is measurement said to disturb the system?

- **Postulate of Measurement:**
  Any device that measures a qubit has two preferred states forming an **orthonormal basis** (e.g., {$|0\rangle$, $|1\rangle$}).
- **Mathematics:**
  If the state is

$$|\psi\rangle = a|0\rangle + b|1\rangle,$$

  then:
  - Probability of outcome $|0\rangle$ = $|a|^2$.
  - Probability of outcome $|1\rangle$ = $|b|^2$.
- **Collapse of State:**
  After measurement, $|\psi\rangle$ **collapses** into the observed eigenstate.
  Example: If $|\psi\rangle$ is measured as $|0\rangle$, then subsequent measurements give $|0\rangle$ with probability 1.
- **Disturbance of System:**
  - Measurement **irreversibly changes** the quantum state.
  - The original amplitudes (a, b) are lost.
  - No further measurement can reveal the initial superposition.
- **Consequences:**
  - Only **1 classical bit** can be extracted per qubit.
  - This disturbance underlies **quantum cryptographic security** (e.g., BB84).
  - It also explains why full quantum state tomography requires many copies of identically prepared qubits.

35. Describe the BB84 quantum key distribution protocol in detail. How does it ensure detection of eavesdropping?

- **Introduction:**
  BB84 (Bennett & Brassard, 1984) is the first **quantum key distribution (QKD)** protocol.
- **Steps:**
  1. **Key Generation by Alice:** She prepares a random bit string.
  2. **Encoding in Qubits:** Each bit is encoded in either the **standard basis** ($0 \rightarrow |\uparrow\rangle$, $1 \rightarrow |\rightarrow\rangle$) or the **Hadamard basis** ($0 \rightarrow |\nearrow\rangle$, $1 \rightarrow |\nwarrow\rangle$).
  3. **Transmission:** Photons (qubits) are sent to Bob via a quantum channel.
  4. **Measurement by Bob:** Bob randomly selects a basis for each photon.
  5. **Basis Reconciliation:** Alice and Bob compare bases (not values) via a classical channel. Only bits where bases match are kept.
  6. **Error Checking:** A subset of bits is compared. If errors exceed threshold, eavesdropping is detected.
- **Eavesdropping Detection (Eve's Attack):**
  - Eve must measure photons before resending them.
  - With 50% chance, she chooses the wrong basis.
  - Wrong measurement alters photon state → Bob's results are wrong with ~25% probability.
  - When Alice and Bob compare sample bits, high error rate reveals Eve's presence.
- **Security:**
  - Based on quantum principles, not computational hardness.
  - The **no-cloning principle** prevents Eve from keeping a copy for later measurement.
  - Even noisy channels can be secured using **error correction** and **privacy amplification**.

BB84 ensures secure key distribution with **built-in eavesdropping detection**, making it a cornerstone of quantum cryptography.

36. Discuss in detail the concept of global phase and relative phase in qubit states. Give mathematical examples to show their physical significance.

- **Global Phase:**
  - If $|\psi\rangle$ is a state, then multiplying by a complex unit factor $e^{i\varphi}$ gives
    $$|\psi'\rangle = e^{i\varphi}|\psi\rangle.$$
  - Both states are physically indistinguishable.
  - Example: $|0\rangle$ and $-|0\rangle$ behave identically under all measurements.
  - Thus, global phase has **no observable effect**.
- **Relative Phase:**
  - The phase difference between components in a superposition is physically important.
  - Example:
    $$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$
  - Although both have equal amplitudes, they are **different states**.
  - Measurement in the Hadamard basis distinguishes them:
    - $|\psi_1\rangle$ collapses to $|\nearrow\rangle$ deterministically.
    - $|\psi_2\rangle$ collapses to $|\nwarrow\rangle$ deterministically.
- **Physical Significance:**
  - Relative phase governs **interference effects**.
  - It is exploited in quantum algorithms (e.g., Grover's, Shor's) and communication protocols.
  - Global phase is irrelevant, but **relative phase enables computational advantage**.

37. Explain with derivation why only one bit of classical information can be extracted from a qubit. Relate your answer to the no-cloning principle.

- **General State:**

  A qubit state is:
  $$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1.$$
- **Measurement Limitation:**
  - Any measurement in $\{|0\rangle, |1\rangle\}$ yields:
    - $|0\rangle$ with probability $|a|^2$.
    - $|1\rangle$ with probability $|b|^2$.
  - Thus, only **one classical outcome** (0 or 1) is revealed.
- **Why not more?**
  - To fully reconstruct (a, b), multiple measurements are needed.
  - But a single qubit collapses after one measurement.
  - Without multiple identical copies, full information is lost.
- **No-Cloning Principle:**
  - Quantum mechanics forbids creating perfect copies of an unknown qubit.
  - Hence, we cannot bypass the measurement limitation by duplication.
- **Conclusion:**
  - Even though qubits encode infinite possible states, measurement yields only **1 classical bit**.
  - This property ensures both the **power and security** of quantum systems.

38. Using photon polarization as an example, explain how quantum mechanics models qubits and their measurement.

- **Physical Representation:**
  - A photon's polarization can represent a qubit.
  - Vertical polarization (|↑⟩) = |0⟩.
  - Horizontal polarization (|→⟩) = |1⟩.
  - Any other angle corresponds to a superposition:

$$|\psi\rangle = a|\uparrow\rangle + b|\rightarrow\rangle.$$

- **Measurement Principle:**
  - A polarizing filter acts as a measurement device.
  - If aligned vertically:
    - Photon passes with probability $|a|^2$.
    - Photon absorbed with probability $|b|^2$.
  - If passes, polarization collapses to |↑⟩.
- **Superposition Example:**
  - 45° photon polarization:

$$|\nearrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\rightarrow\rangle).$$

  - Measurement in standard basis yields |↑⟩ or |→⟩ with equal 50% probability.
- **Implication:**
  - Shows how qubits differ from classical bits.
  - Measurement is **probabilistic** and changes the state.
  - This principle extends to all quantum systems.

39. Explain the exponential growth of quantum state spaces with respect to the number of qubits. How does this differ from classical systems?

- **Classical systems:**
  - An n-bit classical register can store only **one** of 2^n possible values at a time.
  - The total state space = set of possible states, but the system exists in exactly one state.
- **Quantum systems:**
  - An n-qubit register is described by a vector in a 2^n-dimensional Hilbert space.
  - The general state is:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle, \quad \sum |\alpha_i|^2 = 1$$

  - Each coefficient α_i gives the probability amplitude of observing basis state $|i\rangle$.
  - Superposition allows simultaneous encoding of all 2^n possibilities.
- **Difference from classical:**
  - Classical state spaces combine via **direct sum** ($m + n$).
  - Quantum state spaces combine via **tensor product** ($m \times n$), leading to exponential growth.
- **Significance:**
  - This exponential scaling underpins the computational power of quantum computers (e.g., Shor's algorithm, Grover's algorithm).

40. Define and explain entangled states. Discuss their types and applications in quantum information processing.

- **Definition:**

  An entangled state is one that cannot be expressed as a product of individual qubit states.
- **Types:**
  - **2-qubit entangled states (Bell states):**

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

  - **3-qubit entangled states:**
    - GHZ state: $|GHZ\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$.
    - W state: $|W\rangle = (|001\rangle + |010\rangle + |100\rangle)/\sqrt{3}$.
- **Properties:**
  - Measurement outcomes are correlated across qubits.
  - Cannot be described by classical probability distributions.
  - Basis-dependent and decomposition-dependent.
- **Applications:**
  - **Quantum teleportation:** Transmission of unknown quantum states.
  - **Superdense coding:** Sending 2 classical bits with 1 qubit.
  - **Quantum key distribution (E91):** Secure communication guaranteed by entanglement.
  - **Quantum computing:** Resource for speedups in algorithms.
- **Significance:**

  Entanglement is considered the **core resource** of quantum information science.

41. Describe the process of multi-qubit measurement with an example. How does partial measurement affect entangled states?

- **General principle:**
  - A quantum measurement projects the system into one of the basis states, with probabilities given by squared amplitudes.
  - For n qubits, measurement in the computational basis yields one of 2^n outcomes.
- **Example (2-qubit state):**

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$$

  - Measuring the **first qubit**:
    - Outcome 0 → probability = $|a|^2 + |b|^2$, state becomes $\frac{a|00\rangle + b|01\rangle}{\sqrt{|a|^2+|b|^2}}$.
    - Outcome 1 → probability = $|c|^2 + |d|^2$, state becomes $\frac{c|10\rangle + d|11\rangle}{\sqrt{|c|^2+|d|^2}}$.
- **Partial measurement on entangled states:**
  - For Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$:
    - If Alice measures her qubit and gets $|0\rangle$, Bob's qubit collapses to $|0\rangle$.
    - If Alice gets $|1\rangle$, Bob's qubit collapses to $|1\rangle$.
  - This shows **non-local correlations**.

Measurement is probabilistic, collapses the system, and entangled states show correlations even across distance.

42. Explain Ekert's entanglement-based QKD protocol (E91) and compare it with BB84.

- **kert's Protocol (E91):**
  - Proposed by Artur Ekert (1991).
  - Uses **entangled qubits** instead of single qubits.
  - Steps:
    1. Entangled pairs distributed to Alice and Bob.
    2. Each measures qubits in randomly chosen bases.
    3. When their bases match → correlated outcomes form raw key.
    4. Subset compared publicly → check **Bell's inequality**.
    5. If no eavesdropping detected → perform error correction & privacy amplification → final key.
- **Security:**
  - Any eavesdropping breaks entanglement, introducing errors.
  - Violation of Bell's inequality ensures correlations are truly quantum.
- **Comparison with BB84:**

Feature	BB84 (1984)	Ekert E91 (1991)
Resource used	Single qubits in random bases	Entangled pairs (Bell states)
Security principle	No-cloning theorem	Bell's inequality violation
Practicality	Easier to implement, widely used	More difficult, needs entanglement source
Detection of Eve	Increased error rate in key	Correlation disturbance from broken entanglement
Strength	Practical and useful	Stronger theoretical guarantees

Both protocols achieve **unconditional security**, but Ekert's uses **entanglement** as a fundamental resource, while BB84 is more practical today.